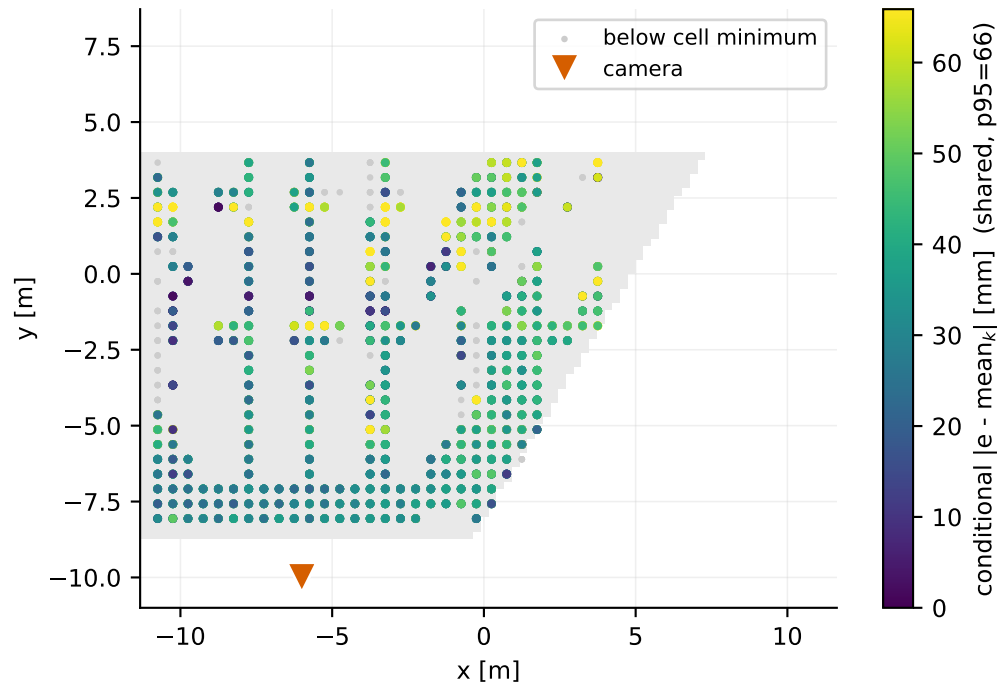
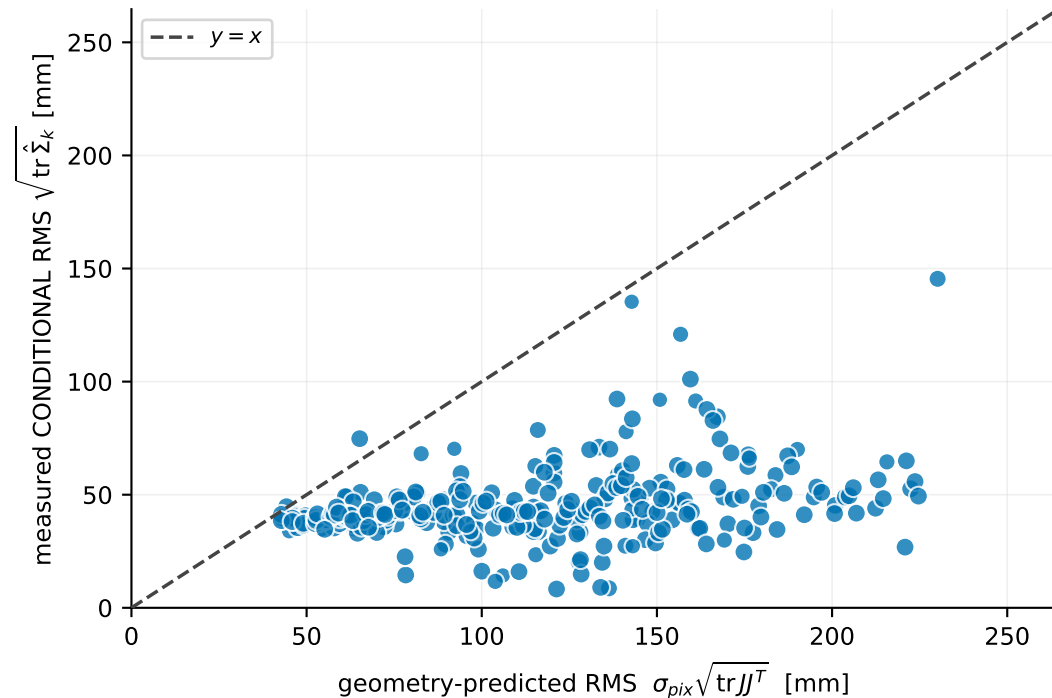


Camera A — deployed-corrected residual

(a) where the robot was actually observed
colour = error AFTER removing the local mean (bias split out)



(b) does geometry predict the local SPREAD?
N = 2556 detections · M = 316 cells (0.5 m, ≥ 5 each)
conditional: R^2 vs identity = -23.89 · cell Spearman = +0.36 · median obs/pred = 0.44
median per-cell BIAS removed first = 47 mm



Point area in (b) scales with detections in the cell. σ_{pix} is fitted on leave-region-out folds, so each cell's prediction is out-of-region. Ground truth positions the dots in (a) and measures the residual; it never enters a projection, Jacobian or covariance. COVERAGE: 14.5 x 11.7 m sampled in two dimensions across 8 headings; image column is decorrelated from range ($|\rho| = 0.26$), so this dataset CAN separate image position from distance. It still cannot separate $\sigma_{max}(J)$ from range: those stay collinear at $|\rho| \sim 0.98$ because for an oblique ground-plane camera the amplification is set by depth, which is what range measures. That is a property of the geometry, not of the sampling.